

SIEMENS EDA

Catapult[®] Synthesis Installation Manual

Software Version 2025.2_1
June 2025

SIEMENS

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Chapter 1: Hardware and Software Requirements

1.1. Operating System Requirements

NOTE: *Catapult products are only available on the Linux platform.*

Catapult supports the Linux operating system (Rocky Linux 8.10). According to the Rocky Linux Wiki, Rocky Linux 8.10 will reach end of life on May 2029. Your system may require one or more operating system patches installed before or after you install Catapult software.

The following Linux windowing systems are supported:

- KDE
- Gnome

1.2. Supported Software and Compilers

Refer to the section “Compatible Compilers and System Software” in the [Catapult Release Notes](#) for detailed information about supported configurations on each platform.

1.3. License Server Requirements

Catapult supports the *saltd* license daemon. The *saltd* daemon is compatible with all prior Catapult releases.

It is highly recommended that you use version 11.19.4.1 (or later) of the *saltd* license daemon (based on the v4.2 Siemens License Server release). Although Catapult only requires *saltd* 11.19.0.1 (or later), the newer version of *saltd* provides some license consumption bug fixes and runs natively on Rocky 9.X. This updated *saltd* license daemon provides immediate benefits plus prepares you for future OS updates.

As shown in the following example, you can determine the version of the current *saltd* daemon using the linux *lmstat* command. If you do not include the *-c* switch, *lmstat* will return the following information for all available license servers.

```
> # lmstat -c <port>@<host>
> $MGC_HOME/bin/lmstat -c 1717@my_lic_server

lmstat - Copyright (c) 1989-2022 Flexera. All Rights Reserved.
Flexible License Manager status on Thu 1/16/2025 13:32

License server status: 1717@my_lic_server
```

```
License file(s) on my_lic_server: /opt/Siemens/ActiveLicenses/cat_hls.lic:
my_lic_server: license server UP (MASTER) v11.19.5
Vendor daemon status (on my_lic_server):
    saltd: UP v11.19.5
    mgcld: UP v11.19.5
```

If you are not currently using the expected *saltd* version, the recommended licensing software can be found at either of the following locations:

- You can download the *Siemens License Server* v4.2 (or later) package, which includes *saltd* v11.19.4.1. The *Siemens License Server* can be downloaded from Siemens SupportCenter.
- The Catapult install tree includes the FlexNet 11.19.5 package (which includes v11.19.5 version of the *saltd* daemon).

For more detailed instructions on using the *Siemens License Server*, refer to `$MGC_HOME/shared/pdffdocs/sw_siemens_license_server_install.pdf`.

Chapter 2: Setting Up Licensing

Catapult products support licensing for node-locked and floating licensing styles with the Siemens Licensing daemon (*saltd*). The Catapult products require authorization codes (licenses) in order to run. Refer to [Step 2: Contact Siemens to Obtain Your Authorization Codes](#) on page 3 for instructions.

The most recent licensing packages can be downloaded from the Siemens SupportCenter site. These packages are compatible with all prior software releases and can be used in an existing Production environment.

NOTE: Licensing must be properly setup prior to the installation of Catapult products.

NOTE: Although Catapult requires version 11.19.0.1 (or later) of the *saltd* license daemon, **it is highly recommended that you use version 11.19.4.1**. Siemens License Server, containing the *saltd* license daemon, is available from within the Catapult installation tree or from SupportCenter by downloading the Siemens License Server v4.2 (or later) package.

Step 1: Determine Your Hostid

Type the following command:

```
$MGC_HOME/bin/lmhostid
```

Step 2: Contact Siemens to Obtain Your Authorization Codes

Please submit a customer support ticket at <http://support.sw.siemens.com> with the following information:

- hostid
- name
- address
- email
- phone
- fax number
- type of license (floating or node-locked)

Step 3: Prepare Your License File

When you receive a new license file, it contains a couple of placeholder values for your license server name and your path to the **saltd** daemon. You must edit the license file and replace these placeholders with actual values. The following sample license file is used to illustrate the procedure.

```
SERVER put_server_name_here 00123F7808DF 1717
DAEMON saltd path_to_saltd
INCREMENT catapultsultra_c mgcld 2025.060 7-jun-2025 2 \
      FDA41B1B814C4B771BA1 VENDOR_STRING=04CFB108 SN=4882439
```

1. Edit the SERVER line:

Identify the system designated as the server.

- a. Check the SERVER line of the provided license file. The third field ("00123F7808DF" in the above example) is the *hostid* of the license server computer. This value must match the *hostid* of the actual node that you use as your license server. Refer to [Step 1: Determine Your Hostid](#) for instructions about obtaining a *hostid*.

The third field of the SERVER line should **never** be edited. If it is incorrect, the licenses need to be regenerated with the correct *hostid*. Editing this field invalidates your licenses.

- b. Prepare the license file for editing. If the license file was sent as an attachment to email, detach the file and save it to a directory which permits you to make edits to the file. If the licensing information is embedded in an email, copy the contents of the email to the file you are permitted to edit. Edit the file to remove any email header information.
- c. Determine the hostname of the license server node. In our example, the server host is a Linux machine accessed through a network. Find its name with the `hostname` command:

```
% hostname
hood
```

The hostname is *hood*.

D. Edit the SERVER line of the license file to reflect the correct hostname. For example:

BEFORE:

```
SERVER put_server_name_here 00123F7808DF 1717
```

AFTER:

```
SERVER hood 00123F7808DF 1717
```

2. DAEMON line:

The DAEMON line shows where the *saltd* license manager software (or vendor daemon) resides.

Edit the DAEMON line to include the path to the *saltd* licensing daemon, as referenced from the license server node. The path to this software varies by tool and must be stated as the absolute path if it is not in the default location. If it is in the default location, only the leaf name is needed. Catapult installs a copy of the *saltd* daemon at:

```
<catapult_install_dir>/Mgc_home/pkgs/FlexNet-11-19-5/Lnx64_x86-64/saltd
```

For the purposes of this example, assume the <catapult_install_dir> is “/cs_home”

```
BEFORE:    DAEMON saltd path_to_saltd
AFTER:    DAEMON saltd /cs_home/Mgc_home/pkgs/FlexNet-11-19-5/Lnx64_x86-64/saltd
```

3. INCREMENT line(s):

The last items to check are the INCREMENT lines. Normally, there should be no need to modify these lines. Notice that each INCREMENT line has several key words and can continue on multiple lines with the ‘\’ used as the line continuation character. INCREMENT lines cannot continue on multiple lines without the line continuation character.

Remove all trailing spaces from the INCREMENT lines. This is sometimes caused by electronic mail formatting and trailing spaces are not allowed within the license file.

At this point your license file should be ready to use. Save it somewhere as a backup copy of the licenses.

Step 4: Install the License File

Copy the license file to a location where the software can access it. For example, the license file is named “*cs_license_host.dat*” could be installed on a network server named “*hood*” at the following location:

```
/net/hood/server/cs_license_host.dat
```

Catapult utilizes the following search order to locate a valid license file. You must use at least one of these methods to point to the license file.

1. -salt_license_server command line switch with Catapult (quoted, semicolon-separated list)
2. -mgls_license_file command line switch with Catapult (colon-separated list)
3. SALT_LICENSE_SERVER environment variable (quoted, semicolon-separated list)
4. MGLS_LICENSE_FILE environment variable (colon-separated list)
5. LM_LICENSE_FILE environment variable (colon-separated list)

If you define multiple servers in the list of licenses, Catapult attempts to locate a valid license in the order specified in the list. You can specify [port@host](#) entries or the path to the license file entries in the list.

NOTE: `LM_LICENSE_FILE` is typically available for any product that uses FlexLM for licensing. `SALT_LICENSE_SERVER` and `MGLS_LICENSE_FILE` are only used by Siemens EDA products.

Step 5: Start the License Server

You should use a log file, especially when starting a new server. The log file provides tracking and troubleshooting information helpful for resolving any future problems. The following list describes how to start the license daemon on a Linux machine:

1. Start the server using the `lmgrd` command:

```
lmgrd -c <path_to_the_license_file> -log <path_to_the_log_file>
```

For example:

```
/cs_home/Mgc_home/bin/lmgrd -c /net/hood/server/cs_license_host.dat  
-log licenses/saltd.log
```

Tip: Ensure that you provide an actual filename for the log file. Unspecified problems may occur if you only specify a directory when using the `-log` switch. You may store the log file anywhere you wish, however storing it in the same directory as the license file makes it easier to cross-check information between the two when troubleshooting.

2. Press the Enter key and the shell prompt displays.
3. Verify that the license server is working by issuing the `lmstat` command:

```
lmstat -a -c <path_to_license_file>
```

For example:

```
/cs_home/Mgc_home/bin/lmstat -a -c /net/hood/server/cs_license_host.dat
```

Chapter 3: Installing Catapult Software

After you verified that licensing is available, you can install the Catapult software. The installation process only installs products with available licenses. The installer can be run in GUI or Batch mode. If the DISPLAY environment variable is set, then the installer uses the Setup GUI otherwise the installer is run in batch mode from the command line.

NOTE: Catapult products are only available on the Linux platform.

3.1. Verifying Environment Variables

Configure the following environment variables before installing Catapult software.

3.1.1. DISPLAY

The DISPLAY environment variable controls whether the Catapult installer is run in GUI or Batch mode. If the DISPLAY environment variable is not set or set to an invalid display name, the Catapult installer will run in batch (command line) mode.

3.1.2. SALT_LICENSE_SERVER, MGLS_LICENSE_SERVER and LM_LICENSE_FILE

Client workstations access licenses from servers via an environment variable or Catapult command line switch. Catapult utilizes the following search path to locate the license file. You must use at least one of these methods to point to the license file.

1. -salt_license_server command line switch with Catapult (quoted, semicolon-separated list)
2. -mgls_license_file command line switch with Catapult (colon-separated list)
3. SALT_LICENSE_SERVER environment variable (quoted, semicolon-separated list)
4. MGLS_LICENSE_FILE environment variable (colon-separated list)
5. LM_LICENSE_FILE environment variable (colon-separated list)

If you define multiple servers in the list, Catapult attempts to locate a valid license in the order specified in the list. You can specify [port@host](#) entries or the path to the license file entries in the variable. When pointing to a license server, the recommended format is [port@host](#).

NOTE: LM_LICENSE_FILE is typically available for any product that uses FlexLM for licensing. SALT_LICENSE_SERVER and MGLS_LICENSE_FILE are only used by Siemens EDA products.

The “port” is the port number through which the license server communicates. The port number is the last field of information on the SERVER line in the license file. In the following example the port number is 1700:

```
SERVER my_lic_server 77446B53 1700
```

The “host” is the hostname of the license server. This can be obtained by looking at the second field of information on the SERVER line in the license file, or by running the `uname -a` command on the license server. In the example SERVER line above, the hostname is `my_lic_server`.

NOTE: The hostname used in the environment variable should be an exact match for the hostname in the license file. If the license file uses the fully qualified domain name, the environment variable should as well; if it does not, the variable should not either.

So, in this example, you could specify the license server order using the following environment variables:

```
setenv SALT_LICENSE_SERVER "1700@my_lic_server;1700@other_lic_server"
```

3.2. Running the Installer

You can download the Linux installer from the Siemens SupportCenter website. The installer will unzip the install tree in the specified directory and then run the Setup Wizard, either in GUI or Batch mode, to configure the install tree based on your available licenses.

To run the installer, follow these steps:

1. Determine whether you want to run in batch or GUI mode

The Setup Wizard can be run in GUI or Batch mode depending on whether your shell has a valid `DISPLAY` environment variable set. If it is set to a valid value, then the Setup Wizard GUI will be used; otherwise the Setup Wizard runs in command line (batch) mode.

2. Execute the installer (`install_linux.bin`) command.
3. Specify the destination directory for the install tree.

The installer program first prompts you for a folder path in which to install the software. You can either accept the default path presented by the installer or enter a different path. If you specify a path that already exists, the program asks if you want to overwrite the existing entries in the folder. Entering “n” will return the installer program to the initial prompt and redisplay the default path. Entering “y” will overwrite the contents of the folder.

4. Click on “Install” to copy the Catapult software into the folder.

After the software has been copied into the installation directory, the installer program launches the Catapult Setup Wizard to guide you through the remainder of the installation process. For detailed information about the setup wizard, see [Using the Catapult Setup Tool](#).

3.2.1. Using the Catapult Setup Tool

The Catapult Setup Wizard configures, uninstalls and patches the installed Catapult software. The wizard is invoked automatically by the installer program when the software is first installed. The installation process is

not complete until after the setup wizard has been run successfully. The Setup Tool can be run in GUI or Batch mode depending on whether your shell has a valid DISPLAY environment variable set.

NOTE: Modifying any files in an installed Catapult tree (renaming, copying, moving, etc.) may invalidate its setup configuration. In that event, the Setup Wizard must be run again to update the configuration. Attempting to invoke any Catapult product from a tree that is not properly configured will automatically invoke the Setup Wizard instead.

NOTE: If the user running the Setup Wizard has write permission to the installation tree, the setup information is saved inside the tree and will apply to all users of the tree. If, however, the user running the Setup Wizard does not have write permission, the setup information is saved in the user's home directory in the file `~/.catapult`, and will apply only to that user.

- If you are running in GUI mode, proceed to “Configuring the Installation using the Setup Wizard GUI” on page 9.
- If you are running in Batch (Command Line) mode, proceed to “Configuring the Installation in Batch Mode” on page 13.

3.2.2. Configuring the Installation using the Setup Wizard GUI

The following steps describe the procedure for configuring the Catapult installation. If you need to configure an existing installation, you can open this wizard using the `catapult -product setup` Linux command.

1. On the welcome page of the wizard, shown in Illustration 1, select “Configure Installation for your computer” and click on the “Next >” button.

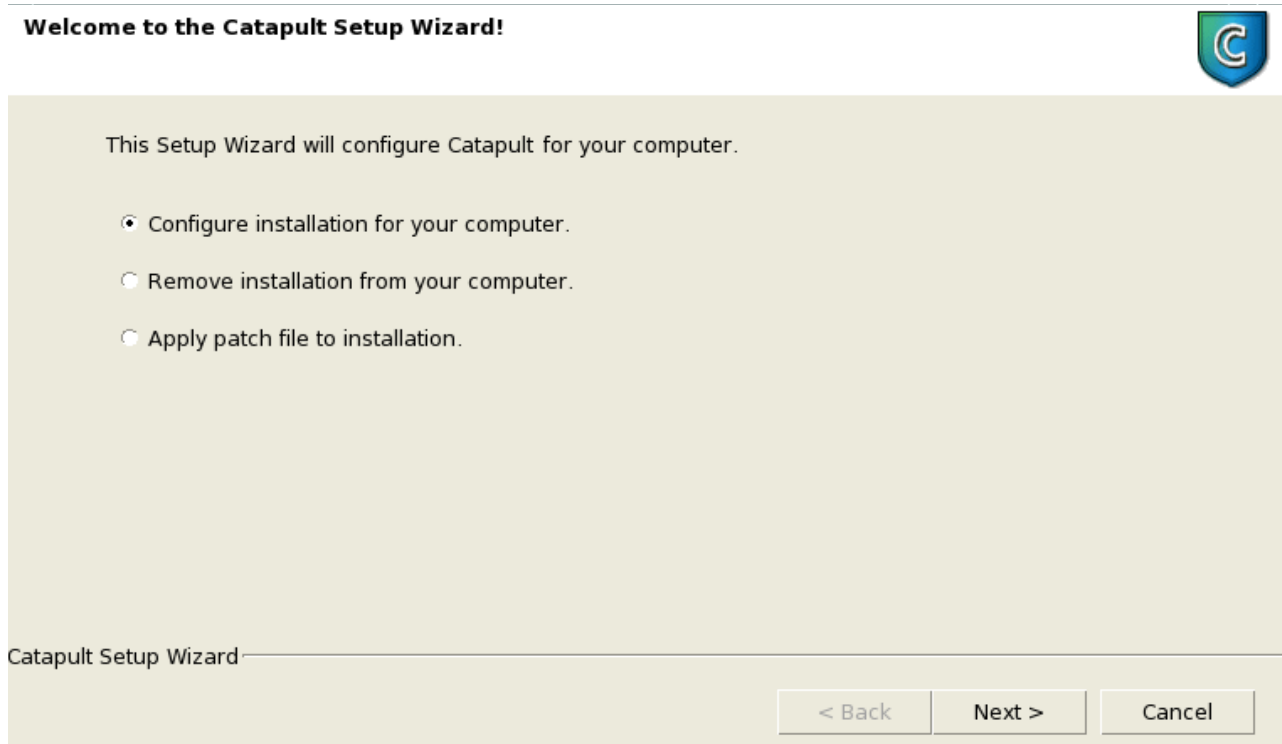


Illustration 1: Welcome Screen

2. The next page, shown in Illustration 2, is the License Verification page. It verifies that license servers are accessible and that proper Catapult licenses are available on the servers. The “License Server” field displays the list of servers specified the MGLS_LICENSE_SERVER or LM_LICENSE_FILE environment variables. If the License Server field is empty, it means neither of the variables were defined (SALT_LICENSE_SERVER is ignored by the installer). If more than one variable is defined, the installer gives precedence to MGLS_LICENSE_FILE. When multiple servers are listed, the server names are separated by colons.

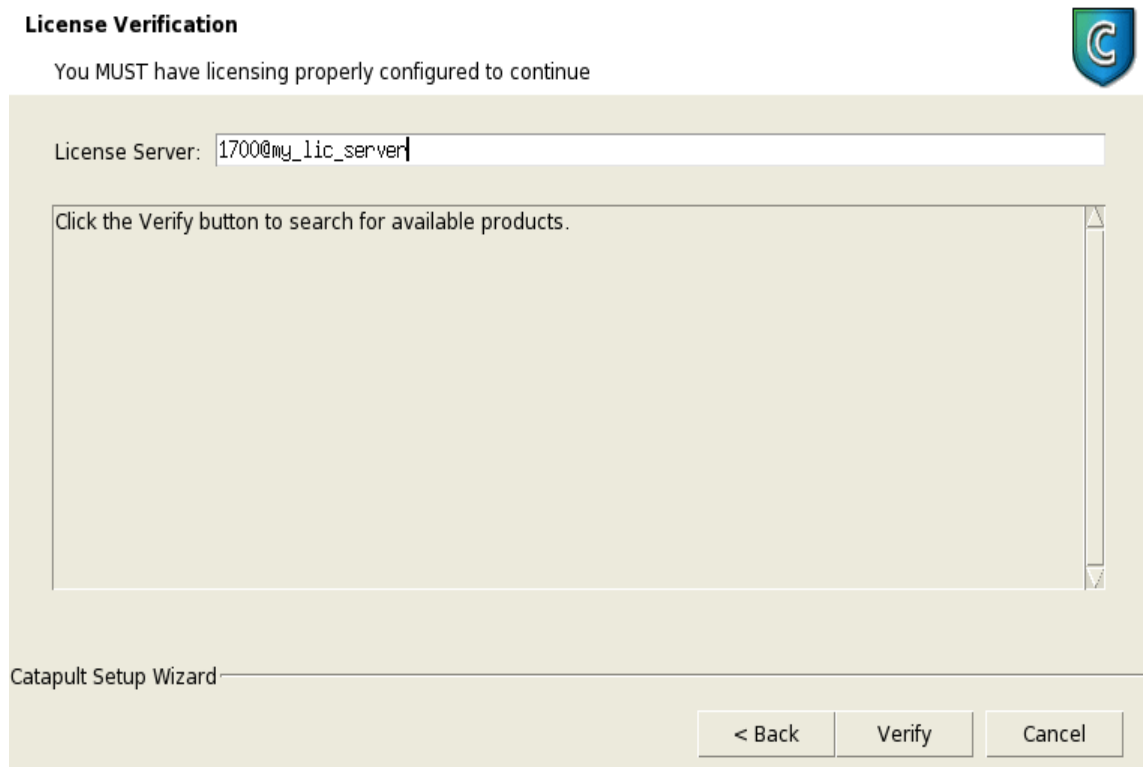


Illustration 2: License Verification Page with License Server List

3. Click the “Verify” button to determine which Catapult licenses are accessible from the license servers. The set of Catapult products you are licensed to install are displayed on the page, as shown in Illustration 3.

If the no license server is accessible, an error message will be displayed in the window. Click the button to launch the licensing management application, “LMTOOLS”.

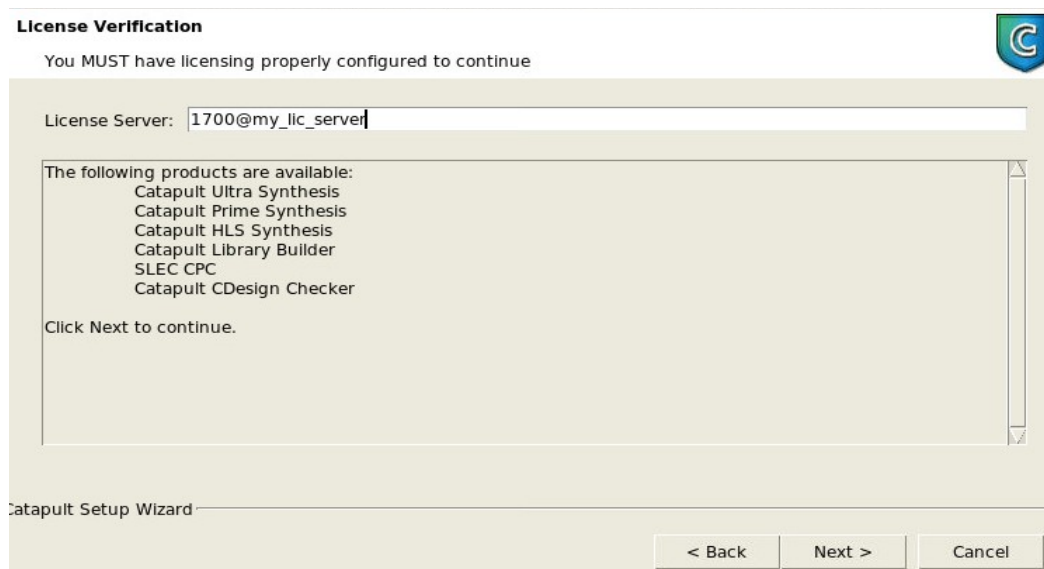


Illustration 3: License Verification Page with List of Licensed Product

- Click the “Next >” button to go to the Product Configuration page, shown in Illustration 4.

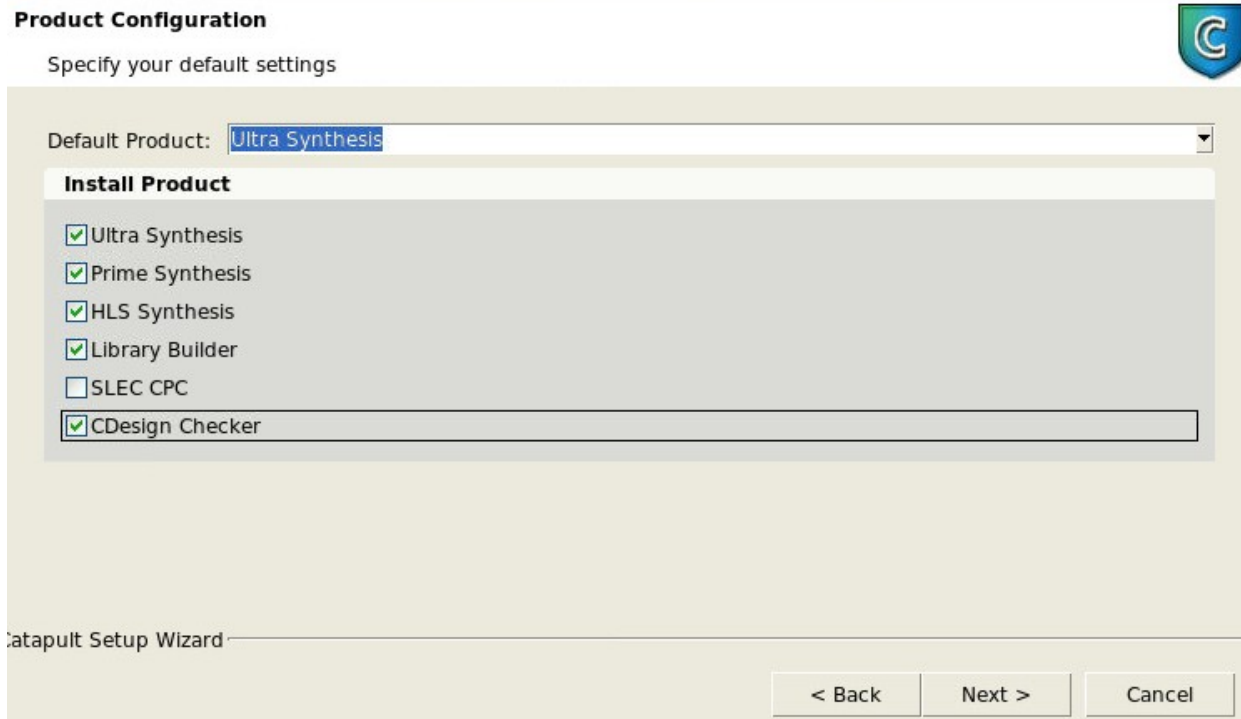


Illustration 4: Product Configuration Page of the Catapult Setup Wizard

This page contains the following fields:

- **Default Product:** This drop-down menu allows you to select one of the products listed in the “Install Product” field to be the default product. The designated default product can be invoked from a shell command line without the “-product” switch.
 - **Install Product:** This field allows you select the set of products to be installed. Click on the check box next to a product name to include or omit it.
5. The next page is the End-User License Agreement. You must accept the license terms in order to continue. Select the acceptance option and click the “Next >” button.

Licensing Agreement



IMPORTANT - USE OF THIS SOFTWARE IS SUBJECT TO LICENSE RESTRICTIONS. Carefully read the following before installing the software:

This software is governed by the terms and conditions found at www.mentor.com/eula. However, such online terms and conditions will not supersede the terms and conditions of any applicable signed agreement governing the use of this software that was entered between you and the licensor from which you are receiving the software.

Having read the entire terms and conditions of the license agreement at www.mentor.com/eula, I agree as authorized representative to accept the terms and conditions of this legally enforceable agreement governing the use of this software.

(Selecting AGREE means you agree to be bound by the applicable terms and conditions.)

Catapult Setup Wizard

6. Click the “Next >” button to go to the Ready to Install page, shown in Illustration 5. Click the “Install” button to install your configuration settings.

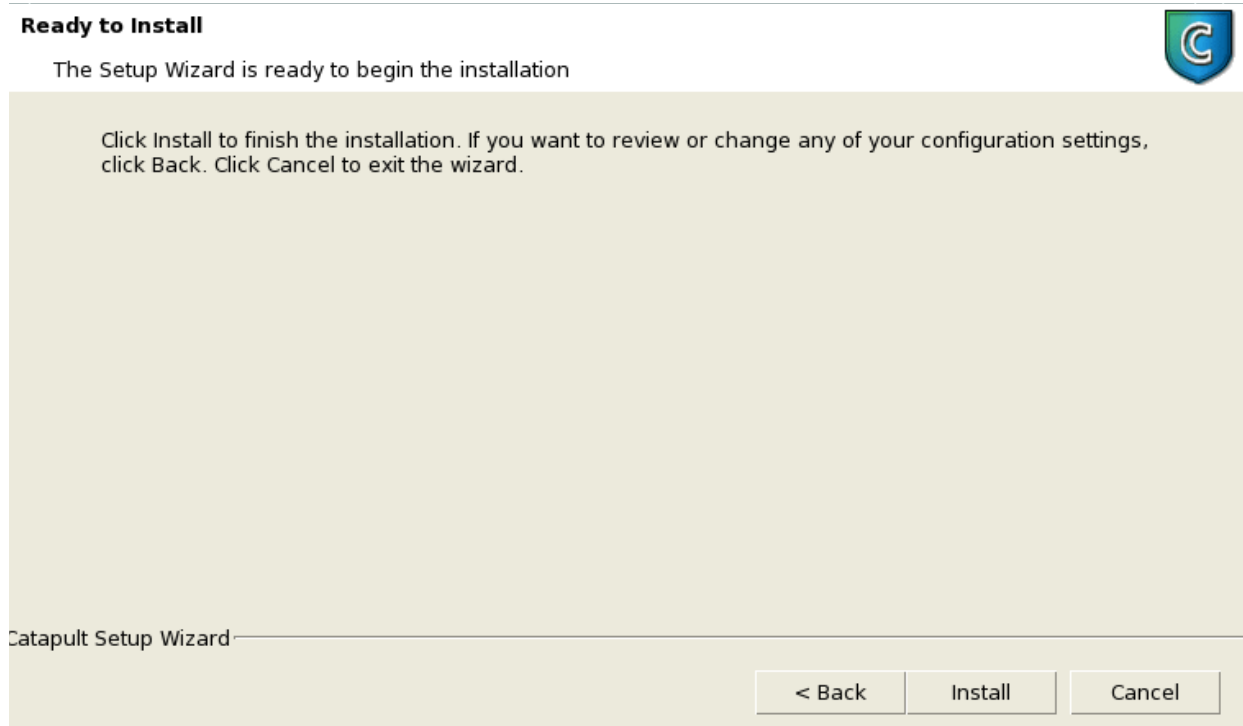


Illustration 5: Ready-to-Install Page of the Catapult Setup Wizard

7. When the install operation is complete, the Finished page appears. Click the “Finish” button to exit the Catapult Setup Wizard.

3.2.3. Configuring the Installation in Batch Mode

The following list describes how to configure the installation from the command line using the product configure and product install commands. At any point, you can view a list of all configuration options by typing product configure –help.

1. Invoke the command line version of the setup wizard using the following Linux command:

```
catapult -shell -product setup
```

2. Specify which products to install.

Catapult has multiple product options. In this step, you will specify which products to configure in a space-separated list. For example, to specify Ultra and Library Builder, you would execute the following command:

```
product configure -products ultra lb
```

The following table lists the available products and their corresponding acronym.

Acronym	Product
all	All Available Products

Acronym	Product
ultra	Catapult Ultra
hls	Catapult HLS
amd	Catapult HLS
lb	Library Builder

3. Specify the default product to invoke

Use the following command to change the default product to invoke:

```
product configure -default ultra
```

The values for the default switch uses the same set of acronyms as the -product switch. The end user can invoke a non-default tool by using the -product switch when they invoke catapult.

4. Specify the path to the license file

The Setup Tool will not install any tool that does not have an available license. In this step, you can specify the license path. This path can be either the [port@host](#) value or the file path name to the license file. For example:

```
product configure -license 1700@my_lic_server
```

5. Specify how to handle the End User License Agreements (EULA)

During the installation process, you must agree to the License Agreement to finish installation of the tool. The Setup Tool allows you to auto-agree to the EULA (*agree*) or prompt the user to accept the EULA during installation (*prompt*).

```
product configure -eulas agree|prompt
```

The following example shows how to setup the installer to auto-agree to the EULA.

```
product configure -eulas agree
```

6. Verify the install configuration by running the product report command. This command will return a summary of the options that will be applied from the default settings (\$HOME/.catapult) and the product configure command. After you are satisfied with the configuration value, you are ready to configure the installation.

7. Configure the installation by executing the *product install* command.

3.3. Invoking Catapult

The Catapult graphical user interface (GUI) can be invoked from a shell command window by entering one of the commands shown in the following table. The commands in the table assume that your PATH variable includes the path to the Catapult executable directory (<install_dir>/Mgc_home/bin).

Catapult invocation commands

Product	Invocation command
Catapult Ultra	catapult -product ultra
Catapult HLS	catapult -product hls
Catapult Formal	catapult -product cformal
Catapult Design Analyzer	catapult -product da
Catapult Library Builder	catapult -product lb
Catapult Setup Wizard	catapult -product setup

NOTE: Catapult products are only available on the linux platform.

To invoke Catapult in non-GUI mode, include the -shell switch. For example:

```
catapult -shell -product ultra
```

Refer to the section “[Catapult](#)” in the Reference > Commands online help for more information about invoking Catapult products.

3.4. Running the Setup Wizard on an existing Install Tree

You can invoke the wizard at any time after installing Catapult with the following shell command:

```
catapult [-shell] -product setup
```

By default, the Setup Wizard in run after the installer. You may have to run the Setup Wizard on an installed tree when you add new licenses. For example, to configure Catapult Ultra as the default invocation for an install tree. By default, Catapult will use the setup Wizard GUI but you can also run the setup tool in batch (shell) mode.

3.5. Removing the Installation

The following steps describe the procedure for uninstalling the Catapult software:

1. On the welcome page of the wizard, shown in [Illustration 1: Welcome Screen](#), select “Remove Installation for your computer” and click on the “Next >” button.

2. The next page asks you to confirm that you want to remove the installation. Select “Yes” and click on the “Next >” button to complete the removal process.

Chapter 4: Third-Party Information

The following HTML page provides information on third-party software that may be included in the Catapult product, including any additional license terms. This document is located at the following location within the Catapult install tree:

```
$MGC_HOME/shared/pkgs/sif_docs/legal/Catapult_OSS.html
```